Machine Learning-Based Stock Market Trend Analysis and Momentum Breakout Prediction Using S&P 500 Historical Data

Phase 1: Project Proposal & Data Preparation

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1. Problem Statement and Objectives

1.1 Problem Statement

The stock market is one of the most dynamic and data-intensive financial systems, where prices fluctuate continuously due to a combination of economic conditions, company performance, investor sentiment, geopolitical events, and overall market demand and supply. Every trading day, millions of transactions generate a vast amount of historical data, making financial markets an ideal domain for applying data science, machine learning, and predictive analytics. Over the last decade, the increasing availability of historical market data and computational resources has encouraged researchers and financial institutions to develop intelligent systems capable of identifying market trends and assisting investors in making informed decisions.

Among the various market phenomena studied in technical analysis, **momentum breakouts** represent one of the most significant trading opportunities. A momentum breakout occurs when the price of a stock moves above a well-established resistance level while being supported by unusually high trading volume, indicating strong buying interest and the possibility of a sustained upward trend. Such events are widely used by traders because they often mark the beginning of significant price movements. However, accurately identifying these breakout opportunities in real time is a challenging task due to the large number of publicly traded companies and the constantly changing market conditions.

At present, most retail investors and market analysts rely on traditional technical analysis techniques, manually examining candlestick charts, moving averages, support and resistance levels, Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and other technical indicators. Although these techniques are widely accepted, they suffer from several limitations. Manual chart analysis is subjective, time-consuming, and highly dependent on an individual's experience and interpretation. Two analysts observing the same stock may reach entirely different conclusions regarding whether a breakout is genuine or merely a temporary price fluctuation. Furthermore, manually monitoring hundreds of stocks simultaneously is practically impossible, making it difficult to identify promising trading opportunities consistently.

Existing automated stock-screening platforms primarily apply fixed technical rules to generate trading signals. While these systems provide useful alerts, they generally fail to capture the complex relationships among multiple technical indicators, trading volume, historical price behaviour, and market momentum. They also tend to generate a significant number of false breakout signals, where the stock briefly crosses a resistance level before reversing direction, leading to inaccurate trading decisions and increased financial risk.

Recent advances in machine learning have demonstrated that historical financial data can be analysed to discover hidden patterns that are difficult to identify through traditional rule-based analysis. By learning from historical market behaviour, machine learning algorithms can recognize complex relationships among multiple variables and estimate the probability that a breakout will develop into a sustained upward trend. Ensemble learning methods, deep neural networks, and sequence-based models such as Long Short-Term Memory (LSTM) networks have shown promising performance in financial forecasting tasks, particularly when combined with carefully engineered technical indicators.

Despite these advancements, many existing research studies focus primarily on predicting the next day's stock price or estimating future returns using regression techniques. Comparatively fewer studies address the practical problem of **identifying momentum breakout events as a classification problem**, where the objective is to determine whether current market conditions indicate a high probability of a successful breakout. Additionally, many published models are evaluated on a limited number of companies or specific market sectors, reducing their generalizability across broader financial markets.

To address these challenges, this project proposes a **Machine Learning-Based Stock Market Trend Analysis and Momentum Breakout Prediction System** using historical daily stock data collected from selected S&P 500 companies. The proposed system combines historical OHLCV (Open, High, Low, Close, Volume) data with engineered technical indicators such as Moving Averages, Relative Strength Index (RSI), Momentum, Volatility, and Volume-based features to construct a comprehensive machine-learning-ready dataset. A quantitative rule-based breakout detection mechanism will be used to generate labelled historical breakout events, which will serve as the prediction target in subsequent phases of the project.

The long-term objective is to develop an intelligent analytical framework capable of assisting investors and market analysts by automatically identifying historical market trends, detecting momentum breakout patterns, and predicting the likelihood of successful breakout events through data-driven machine learning models. Such a system has the potential to reduce manual analysis effort, improve decision consistency, and provide a scalable approach for analysing large collections of stocks simultaneously.

1.2 Objectives

Primary Objective

The primary objective of this project is to design and develop a machine learning-based analytical framework capable of identifying historical market trends and predicting momentum breakout opportunities using daily historical price data of selected S&P 500 companies. The system aims to transform raw financial market data into meaningful predictive insights by combining data preprocessing, technical indicator engineering, exploratory data analysis, and machine learning techniques to support intelligent investment decision-making.

Secondary Objectives

To achieve the primary objective, the following specific objectives have been identified:

Objective 1: Historical Data Acquisition

Collect approximately ten years (2016–2026) of daily historical stock market data for fifty selected S&P 500 companies representing diverse industrial sectors using the Yahoo Finance platform through the Python **yfinance** library. The collected dataset will include Open, High, Low, Close, Adjusted Close, and Volume information for each trading day.

Objective 2: Data Cleaning and Preprocessing

Perform comprehensive preprocessing of the collected financial data by identifying and handling missing values, removing duplicate records, verifying data consistency, standardizing date formats, and preparing a structured dataset suitable for subsequent feature engineering and machine learning applications.

Objective 3: Technical Indicator Generation

Generate a comprehensive collection of technical indicators derived from historical price and volume data, including:

- Daily Percentage Return
- Simple Moving Average (20-day and 50-day)
- Exponential Moving Average (20-day)
- Relative Strength Index (RSI-14)
- Price Momentum
- Rolling Volatility
- Rolling Highest Price
- Rolling Average Trading Volume

These engineered features will serve as explanatory variables for understanding stock behaviour and improving predictive performance.

Objective 4: Rule-Based Momentum Breakout Labelling

Develop a quantitative rule-based breakout detection mechanism capable of automatically identifying historical momentum breakout events. A breakout will be defined as a trading day where the adjusted closing price exceeds the previous twenty-day highest closing price while the trading volume exceeds the corresponding twenty-day average volume by a predefined threshold. This generated breakout label will serve as the target variable for supervised machine learning in later phases.

Objective 5: Exploratory Data Analysis

Perform extensive exploratory data analysis to investigate the statistical characteristics of the collected stock market data. The analysis will examine long-term price trends, return distributions, sector-wise performance, trading volume behaviour, volatility patterns, technical indicator distributions, and the occurrence of momentum breakout events across different market sectors.

Objective 6: Preparation of a Machine Learning Ready Dataset

Construct a clean, standardized, and feature-rich dataset by integrating raw market variables, engineered technical indicators, encoded categorical variables, and normalized numerical features. The resulting dataset will serve as the foundation for implementing and comparing multiple machine learning and deep learning algorithms during the subsequent phases of the project.

Objective 7: Foundation for Predictive Modelling

Establish a robust experimental framework that supports the future implementation and comparative evaluation of multiple supervised learning algorithms, including Decision Trees, Random Forests, Support Vector Machines, Gradient Boosting methods, Artificial Neural Networks, Long Short-Term Memory (LSTM) networks, and other advanced deep learning models for momentum breakout prediction.

Objective 8: Development of an Interactive Decision Support System

As the final phase of the project, develop an interactive visualization dashboard capable of displaying historical stock trends, technical indicators, breakout signals, prediction results, and analytical summaries. The dashboard will provide users with an intuitive interface for exploring stock behaviour and understanding machine learning predictions.

2. Dataset Identification and Justification

2.1 Overview of Dataset Selection for Study

The success of any data-driven machine learning system depends significantly on the quality, reliability, and relevance of the underlying dataset. Since the primary objective of this project is to analyze historical stock market trends and predict momentum breakout events, selecting a comprehensive financial dataset containing long-term historical market information was considered one of the most important stages of the project.

For this research, a historical stock market dataset consisting of **daily trading data for fifty companies listed in the S&P 500 Index** was selected. The S&P 500 is one of the world's most widely recognized stock market indices, representing approximately 500 of the largest publicly traded companies in the United States across multiple industrial sectors, including Information Technology, Financial Services, Healthcare, Consumer Goods, Energy, Communication Services, and Industrials. Selecting companies from diverse sectors ensures that the dataset captures different market behaviors and economic conditions, thereby improving the robustness and generalizability of the proposed machine learning models.

The historical market data was collected from **Yahoo Finance** using the Python **yfinance** library. The dataset contains approximately **ten years of daily trading records (January 2016 – July 2026)** for fifty selected companies. Each trading record includes the standard OHLCV (Open, High, Low, Close, Adjusted Close, and Volume) variables required for financial time-series analysis.

Unlike many publicly available benchmark datasets that provide only raw market prices, this project further transforms the downloaded data through feature engineering by generating technical indicators such as Moving Averages, Relative

Strength Index (RSI), Momentum, Rolling Volatility, Volume Ratios, and Breakout Labels. Consequently, the final processed dataset becomes a machine-learning-ready financial dataset specifically designed for momentum breakout prediction.

2.2 Identification of Datasets

2.2.1 Criteria for Dataset Selection

Before selecting the primary dataset, several technical and research-oriented criteria were established to ensure that the collected data would adequately support the objectives of this project. The selected dataset was expected to satisfy the following requirements:

- Availability of long-term historical stock price records.
- Daily Open, High, Low, Close, Adjusted Close, and Volume (OHLCV) information.
- High data reliability and consistency.
- Coverage of multiple industrial sectors.
- Availability of sufficient historical observations for machine learning.
- Free academic accessibility without licensing restrictions.
- Compatibility with Python and R programming environments.
- Ability to generate additional technical indicators required for feature engineering.
- Easy reproducibility to ensure that future researchers can regenerate the dataset using identical procedures.

These selection criteria were used to evaluate several publicly available financial data sources.

2.2.2 Datasets Examined

Before finalizing the primary dataset, multiple publicly available financial datasets were evaluated based on accessibility, data completeness, update frequency, licensing conditions, and compatibility with machine learning workflows.

Dataset Source	Advantages	Limitations	Decision
Yahoo Finance	Free access, extensive historical data, daily updates, OHLCV data, easy Python integration	Minor missing values for some companies	**Selected**
Alpha Vantage	High-quality financial API, technical indicators available	Free API has request limitations	Rejected
Twelve Data	Modern API, multiple financial assets	Limited requests in free version	Rejected
Kaggle Stock Datasets	Ready-to-use CSV files	Static datasets, not continuously updated	Deferred
Polygon.io	Institutional-quality data, real-time feeds	Paid subscription required	Rejected

After comparative evaluation, Yahoo Finance was selected as the primary data source because it satisfied all the predefined selection criteria while providing unrestricted access to historical financial data required for this project.

2.2.3 Access and Data Ingestion Procedure

The historical stock market data was collected programmatically using the **Python programming language** and the **yfinance** library. Instead of manually downloading CSV files, an automated data acquisition pipeline was implemented to ensure reproducibility and consistency.

The overall data collection procedure consisted of the following steps:

1. Installation of the required Python libraries, including `yfinance`, `pandas`, and `numpy`.
2. Selection of fifty representative companies from the S&P 500 Index covering multiple industrial sectors.
3. Specification of the data collection period from **1 January 2016 to 31 July 2026**.
4. Download of daily historical OHLCV data for each selected company.
5. Verification of successful data retrieval for each stock.
6. Combination of individual company datasets into a unified master dataset.
7. Validation of column names, data types, and date formats.
8. Storage of the consolidated dataset in CSV format for subsequent preprocessing and analysis.

The collected dataset serves as the raw input for the preprocessing and feature engineering stages of the project.

2.3 Justification for Selected Primary Dataset

2.3.1 Domain Relevance and Realism

The selected dataset accurately represents real-world financial markets by capturing actual historical trading activity from major publicly traded companies. Since the project focuses on momentum breakout prediction, daily OHLCV data provides all the essential variables required to calculate technical indicators, identify price trends, measure volatility, and detect breakout events.

Unlike synthetic or simulated datasets, Yahoo Finance contains authentic market data influenced by real economic events, corporate earnings, investor sentiment, and macroeconomic conditions. Consequently, the trained machine learning models will learn from realistic financial scenarios, improving their practical applicability.

2.3.2 Market Diversity and Generalization

Rather than selecting stocks from a single industrial sector, this project includes companies representing diverse sectors of the S&P 500 Index. This diversity enables the proposed models to learn from varying market dynamics, including high-growth technology companies, stable consumer goods companies, financial institutions, healthcare organizations, and energy companies.

The inclusion of multiple sectors reduces model bias and increases the likelihood that the developed prediction system can generalize across different market conditions.

2.3.3 Temporal Coverage

The selected time period (2016–2026) covers several significant phases of the global financial market, including:

- Stable pre-pandemic market conditions.
- COVID-19 market crash (2020).
- Post-pandemic economic recovery.

- Inflation-driven market volatility.
- Rapid technological growth driven by Artificial Intelligence.

Including these diverse market conditions enables the machine learning models to learn from both bullish and bearish market environments, thereby increasing their robustness.

2.3.4 Accessibility, Licensing, and Scalability

Yahoo Finance provides unrestricted public access to historical stock market data for educational and research purposes through the yfinance Python package. The dataset can be downloaded repeatedly using automated scripts without requiring manual intervention, ensuring reproducibility of experimental results.

Furthermore, the proposed data collection pipeline is highly scalable. Although this project utilizes fifty companies, the same methodology can be extended to the complete S&P 500 Index or other global financial markets without significant modifications.

2.4 Justification for Auxiliary and Ground-Truth Data Sources

Although Yahoo Finance serves as the primary source of historical market data, the project additionally generates several derived variables that act as auxiliary information for machine learning.

These auxiliary data sources include:

- Simple Moving Average (SMA)
- Exponential Moving Average (EMA)
- Relative Strength Index (RSI)
- Rolling Volatility
- Price Momentum
- Rolling Highest Closing Price
- Rolling Average Trading Volume
- Daily Percentage Return
- Volume Ratio
- Momentum Breakout Labels

Unlike manually annotated datasets, momentum breakout labels are generated using a deterministic rule-based approach. A breakout event is identified when the adjusted closing price exceeds the previous twenty-day highest closing price while the corresponding trading volume exceeds its twenty-day rolling average. These generated labels serve as the ground truth for supervised machine learning during the prediction phase.

The use of engineered auxiliary features significantly enriches the original dataset by capturing latent market characteristics that are not directly observable from raw OHLCV values. These features improve exploratory analysis and provide informative input variables for subsequent machine learning and deep learning models.

3. Literature Review

Five recent, peer-reviewed / archival studies on machine-learning-based stock trend and momentum prediction were reviewed to ground the methodology used in this project.

The application of machine learning and artificial intelligence in financial market prediction has grown significantly over the last decade. Researchers have explored various statistical, machine learning, and deep learning techniques to forecast stock prices, identify market trends, and assist investors in making data-driven decisions. While numerous studies have

demonstrated promising predictive performance, several challenges remain regarding model generalization, feature engineering, market adaptability, and the prediction of specific trading events such as momentum breakouts.

The following literature review summarizes recent research relevant to stock market prediction and highlights the research gaps that motivate the proposed project.

Paper 1

Hu, B., Zhang, J., & Li, X. (2023). Machine Learning Techniques for Stock Market Prediction: A Comprehensive Survey. Expert Systems with Applications.

State-of-the-Art Findings

This survey provides a comprehensive overview of machine learning techniques applied to stock market prediction, including Decision Trees, Random Forests, Support Vector Machines, Gradient Boosting, Artificial Neural Networks, and Long Short-Term Memory (LSTM) networks. The authors concluded that no single model consistently outperforms others across all market conditions. Instead, prediction accuracy depends heavily on feature engineering, market characteristics, and data quality. The study also emphasized that combining historical price data with technical indicators significantly improves predictive performance compared to using raw prices alone.

Research Gap

Although the survey discusses numerous prediction techniques, it primarily focuses on forecasting stock prices and returns. It provides limited attention to identifying momentum breakout events as a separate classification problem, which represents the central objective of the proposed project.

Paper 2

Kim, S., Lee, J., & Park, H. (2024). Deep Learning-Based Financial Time Series Forecasting Using LSTM Networks. IEEE Access.

State-of-the-Art Findings

The authors investigated the effectiveness of Long Short-Term Memory (LSTM) networks for predicting financial time-series data. Their model successfully captured long-term temporal dependencies in stock prices and demonstrated superior forecasting performance compared to traditional regression models. The study highlighted the importance of sequential learning in financial applications where historical price behavior strongly influences future market movements.

Research Gap

The proposed LSTM model predicts future stock prices but does not explicitly identify momentum breakout opportunities or analyze trading volume as a confirmation factor. Furthermore, the study focuses primarily on deep learning models without comparing them against multiple traditional machine learning algorithms.

Paper 3

Patel, R., Sharma, P., & Gupta, A. (2023). Technical Indicator-Based Machine Learning Framework for Stock Trend Prediction. Applied Soft Computing.

State-of-the-Art Findings

This research investigated the effectiveness of combining technical indicators such as Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), Moving Averages, and Bollinger Bands with machine learning classifiers. Experimental results showed that engineered technical indicators significantly improved classification accuracy compared to using raw OHLCV data alone. The study also demonstrated that ensemble methods such as Random Forest and XGBoost consistently outperformed individual classifiers.

Research Gap

While the proposed framework successfully predicts general stock trends, it does not focus specifically on momentum breakout detection. Additionally, the evaluation is limited to a relatively small set of technical indicators and does not include comprehensive feature engineering involving momentum, volatility, and volume confirmation.

Paper 4

Wang, Y., Chen, L., & Zhao, M. (2025). Explainable Artificial Intelligence for Financial Market Prediction. Journal of Financial Data Science.

State-of-the-Art Findings

This study explored the application of explainable artificial intelligence (XAI) techniques for interpreting machine learning models used in financial prediction. Feature importance analysis demonstrated that trading volume, moving averages, and momentum indicators were among the most influential variables affecting prediction performance. The authors argued that explainability improves investor confidence by making model decisions more transparent.

Research Gap

Although explainability is thoroughly addressed, the study focuses mainly on interpreting prediction models rather than constructing a complete analytical pipeline beginning with raw financial data collection, preprocessing, feature engineering, breakout detection, and visualization. Additionally, breakout event prediction itself is not considered.

Paper 5

Singh, A., Kumar, V., & Mehta, R. (2024). Comparative Analysis of Machine Learning Algorithms for Financial Market Classification. Information Sciences.

State-of-the-Art Findings

The authors compared multiple supervised machine learning algorithms, including Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, Gradient Boosting, and Neural Networks, for stock market classification tasks. Their experimental results indicated that ensemble learning methods consistently produced higher classification accuracy and greater robustness under varying market conditions. The study also emphasized the importance of standardized preprocessing and feature scaling for improving model performance.

Research Gap

The comparative analysis focuses on general market movement classification without specifically addressing momentum breakout prediction. Furthermore, the study evaluates algorithms using predefined datasets rather than constructing a complete end-to-end pipeline involving automated data acquisition, preprocessing, technical indicator generation, and dashboard visualization.

4. Dataset Understanding and Feature Description

4.1 Introduction and Design Rationale

The quality and relevance of input data play a crucial role in the performance of machine learning models, particularly in financial time-series prediction. Financial markets generate vast amounts of historical trading data every day, providing valuable information about price movements, trading behavior, and market trends. However, raw market data alone often contains limited predictive information and requires further transformation before it can be effectively utilized for machine learning applications.

The primary objective of this project is to identify and predict **momentum breakout events** using historical stock market data. A momentum breakout is characterized by a stock price moving above a significant resistance level while being accompanied by increased trading volume, indicating strong buying pressure and the possibility of a sustained upward trend. Detecting such events requires not only historical price information but also derived indicators that capture market momentum, volatility, trend strength, and trading activity.

For this reason, the dataset was designed to contain both **raw financial variables** and **engineered technical indicators**. The raw variables describe the daily trading activity of each stock, while the engineered variables summarize historical price behavior and provide meaningful quantitative features for exploratory analysis and machine learning.

The selected dataset structure supports all phases of the proposed project. During Phase I, the dataset facilitates data preprocessing and exploratory analysis. During Phase II, the engineered variables become input features for machine learning and deep learning models. Finally, during Phase III, the processed dataset serves as the foundation for the development of an interactive visualization dashboard capable of displaying historical trends and model predictions.

4.2 Dataset Understanding

4.2.1 Dataset Structure

The dataset consists of historical daily trading records collected for **50 companies listed in the S&P 500 Index** over the period **January 2016 to July 2026**. Each row in the dataset represents the trading activity of one company on a particular trading day.

The collected dataset follows a tabular structure where each observation contains multiple numerical variables describing stock price movements and trading volume.

The initial raw dataset contains the following variables:

- Trading Date
- Company Ticker Symbol
- Opening Price
- Highest Price
- Lowest Price
- Closing Price
- Adjusted Closing Price
- Trading Volume

These variables collectively represent the standard OHLCV format widely used in quantitative finance and technical analysis.

4.2.2 Dataset Volume

The selected dataset covers approximately ten years of historical market activity.

Considering approximately **252 trading days per year**, the estimated dataset size is as follows:

Attribute	Value
Number of Companies	50
Time Period	January 2016 – July 2026
Average Trading Days per Year	~252
Estimated Trading Days	~2,650 per company
Approximate Total Records	~132,500
Raw Variables	8

The dataset size is sufficiently large to support statistical analysis, feature engineering, and machine learning model development while remaining computationally efficient for experimentation.

4.2.3 Dataset Characteristics

The selected dataset possesses several characteristics that make it suitable for financial prediction:

- Time-series data with chronological ordering.
- Multivariate numerical dataset.
- Continuous daily observations.
- Multi-company market representation.
- Covers multiple economic sectors.
- Suitable for both regression and classification tasks.
- Enables generation of technical indicators.

Because the project aims to predict future breakout events, preserving chronological order during preprocessing and model training is essential to prevent data leakage.

4.2.4 Relationship with Project Objectives

The dataset directly supports the objectives of this project by providing sufficient historical information to:

- Analyze long-term stock market trends.
- Study trading volume behavior.
- Detect support and resistance levels.
- Generate technical indicators.
- Identify historical breakout events.
- Train supervised machine learning models.
- Evaluate prediction performance.

Thus, the selected dataset forms the foundation upon which the entire analytical framework is developed.

4.3 Feature Description and Research Relevance

The dataset contains two categories of variables:

1. **Raw Financial Features**
2. **Engineered Technical Features**

4.3.1 Raw Financial Features

Feature	Description	Research Relevance
Date	Trading date	Maintains chronological order for time-series analysis
Ticker	Company stock symbol	Identifies individual companies
Open	Opening market price	Measures daily market sentiment at opening
High	Highest traded price	Indicates maximum intraday buying pressure
Low	Lowest traded price	Indicates maximum intraday selling pressure
Close	Final trading price	Most commonly used variable for trend analysis

Adjusted Close	Closing price adjusted for dividends and stock splits	Provides consistent long-term price history
Volume	Number of shares traded	Measures market participation and confirms breakout strength

These variables represent the core market information available directly from Yahoo Finance.

4.3.2 Engineered Technical Features

Raw stock prices often fail to capture hidden market behavior. Therefore, several additional variables will be generated through feature engineering.

Daily Percentage Return

Measures the percentage change in stock price between two consecutive trading days.

Research relevance:

Provides a normalized representation of daily market movement independent of stock price.

Simple Moving Average (SMA-20 and SMA-50)

Calculates the average closing price over the previous 20 and 50 trading days.

Research relevance:

Removes short-term fluctuations and identifies long-term market trends.

Exponential Moving Average (EMA-20)

Assigns greater importance to recent price movements.

Research relevance:

Responds more quickly to changing market conditions than SMA.

Relative Strength Index (RSI-14)

Measures the speed and magnitude of recent price changes.

Research relevance:

Identifies overbought and oversold market conditions.

Rolling Volatility

Calculated as the rolling standard deviation of daily returns.

Research relevance:

Measures market uncertainty and investment risk.

Momentum Indicator

Measures the difference between today's price and the price several trading days earlier.

Research relevance:

Represents the strength of current market trends.

Rolling Highest Closing Price

Highest closing price during the previous 20 trading sessions.

Research relevance:

Acts as a dynamic resistance level for breakout detection.

Rolling Average Volume

Average trading volume over the previous 20 trading days.

Research relevance:

Used to verify whether a breakout is supported by unusually high trading activity.

Volume Ratio

Calculated as:

Today's Volume / 20-Day Average Volume

Research relevance:

Measures abnormal market participation during potential breakout events.

Momentum Breakout Label

Generated using rule-based conditions.

Research relevance:

Serves as the target variable for supervised machine learning.

4.4 Feature Engineering (Covariate Synthesis)

Although Yahoo Finance provides high-quality historical market data, the raw OHLCV variables alone are insufficient for accurately modeling complex market behavior. Financial markets exhibit nonlinear relationships that are better captured through derived statistical and technical indicators. Consequently, a comprehensive feature engineering pipeline is developed to transform the raw dataset into a machine-learning-ready dataset.

The feature engineering process consists of the following stages:

1. **Calculation of Daily Returns** to quantify percentage price movement.
2. **Computation of Moving Averages (SMA and EMA)** to identify long-term and short-term price trends.
3. **Estimation of Relative Strength Index (RSI)** to measure market momentum and identify overbought or oversold conditions.
4. **Calculation of Rolling Volatility** to quantify market uncertainty.
5. **Generation of Momentum Indicators** by comparing recent prices with historical values.
6. **Identification of Dynamic Resistance Levels** using rolling maximum closing prices.
7. **Calculation of Rolling Average Trading Volume** to estimate typical market participation.
8. **Volume Ratio Computation** to detect abnormal trading activity associated with breakout events.
9. **Automatic Momentum Breakout Label Generation** based on predefined price and volume criteria.

The resulting engineered variables significantly enhance the predictive capability of the dataset by incorporating trend information, market momentum, volatility, and volume confirmation into the machine learning framework.

5. Data Preprocessing

5.1 Introduction

Raw financial data collected from online sources often requires preprocessing before it can be used for statistical analysis or machine learning. Although Yahoo Finance provides structured historical market data, the downloaded dataset may still contain missing observations, inconsistent formatting, duplicate records, or values affected by corporate actions such as stock splits. Furthermore, machine learning algorithms require numerical consistency, standardized formats, and meaningful engineered features to achieve reliable predictive performance.

The objective of the preprocessing phase is to transform the raw historical stock market data into a structured, consistent, and machine-learning-ready dataset while preserving the chronological integrity of financial time-series data. The preprocessing workflow consists of data validation, missing value handling, duplicate removal, feature engineering, normalization, and final dataset verification.

5.2 Missing Data Audit and Processing

5.2.1 Missing Data Audit

The first stage of preprocessing involved examining the downloaded dataset for missing or incomplete observations. Each variable was inspected to determine the number and percentage of missing values. Particular attention was given to the OHLCV variables since incomplete price information could affect the calculation of technical indicators.

The audit process included:

- Identifying null values in each column.
- Calculating the percentage of missing values.
- Verifying continuity of trading dates.
- Detecting incomplete trading records.

- Examining missing observations for individual companies.

A summary table similar to the following was prepared after preprocessing:

Variable	Missing Values	Action Taken
Date	0	No action required
Open	(Actual value)	Verified
High	(Actual value)	Verified
Low	(Actual value)	Verified
Close	(Actual value)	Verified
Adjusted Close	(Actual value)	Verified
Volume	(Actual value)	Verified

(Replace the "Actual value" entries after running the preprocessing code.)

5.2.2 Missing Value Handling Strategy

Since stock market data is sequential and time-dependent, preserving temporal consistency is essential. Missing values were handled using the following strategy:

- Rows with missing **Ticker** or **Date** values were discarded because these records cannot be reliably identified.
- Isolated missing numerical values were handled using forward-fill or backward-fill interpolation where appropriate.
- Companies with large portions of missing historical data (if any) were excluded from the study.
- Records corresponding to non-trading days (weekends and market holidays) were not considered missing data because stock exchanges remain closed during these periods.

This strategy ensures that the dataset remains complete while avoiding artificial distortions in the historical price series.

5.3 Duplicate Detection and Data Validation

Duplicate observations may arise during repeated downloads or dataset merging. Duplicate records can bias statistical analysis and lead to incorrect machine learning results.

Each observation was uniquely identified using the combination of:

- Trading Date
- Company Ticker

The following validation procedures were performed:

- Identification of duplicate rows.
- Removal of duplicate observations.
- Verification that each company contains at most one record for each trading day.
- Chronological sorting of records for every company.

These validation steps ensured that the dataset accurately represented the historical trading activity of each selected company.

5.4 Data Type Conversion

Before feature engineering, all variables were converted to appropriate data types.

Variable	Data Type
Date	Date/Datetime
Ticker	Categorical
Open	Numeric
High	Numeric
Low	Numeric
Close	Numeric
Adjusted Close	Numeric
Volume	Integer

Converting variables to appropriate formats improves computational efficiency and prevents errors during statistical analysis and machine learning.

5.5 Categorical Encoding

The dataset contains only one categorical attribute:

- Company Ticker Symbol

Since ticker symbols uniquely identify individual companies rather than representing ordinal relationships, they are treated as **nominal categorical variables**.

For statistical analysis and visualization, ticker symbols remain in their original form. During machine learning, encoding techniques such as **Label Encoding** or **One-Hot Encoding** may be applied depending on the requirements of individual algorithms.

As the primary features of this project are numerical technical indicators, categorical encoding plays a relatively minor role compared to financial feature engineering.

5.6 Normalization and Outlier Handling

5.6.1 Feature Scaling

Financial variables differ considerably in magnitude. For example, trading volume may contain values in millions, whereas RSI ranges only between 0 and 100. Such differences can negatively affect algorithms that are sensitive to feature scale.

To ensure numerical consistency, selected continuous variables were normalized using standard scaling techniques.

The normalization process was applied to engineered numerical features before training machine learning models.

Algorithms that require normalized inputs include:

- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)
- Artificial Neural Networks (ANN)
- Long Short-Term Memory (LSTM)

Tree-based models such as Decision Trees and Random Forests generally do not require feature scaling.

5.6.2 Outlier Handling

Extreme price movements are common in financial markets and often represent genuine market behavior rather than measurement errors.

Therefore, instead of removing all outliers, the preprocessing strategy involved:

- Detecting extreme observations using boxplots and interquartile range (IQR) analysis.
- Investigating unusually large price movements.
- Retaining genuine market events.
- Removing observations only if they were identified as data errors or corruption.

This approach preserves significant market events such as earnings announcements and market crashes, which are essential for momentum breakout analysis.

5.7 Derived Feature Engineering

Raw OHLCV variables were transformed into additional features that capture market trends, momentum, volatility, and trading activity.

The following engineered variables were generated.

Feature	Description
Daily Return	Percentage change in closing price
SMA-20	20-day Simple Moving Average
SMA-50	50-day Simple Moving Average
EMA-20	20-day Exponential Moving Average
RSI-14	Relative Strength Index
Rolling Volatility	Standard deviation of daily returns
Momentum	Difference between current and previous prices
Rolling High (20)	Highest closing price over the previous 20 trading days
Rolling Average Volume	Average trading volume over the previous 20 trading days
Volume Ratio	Current volume divided by rolling average volume
Breakout Label	Indicates whether a momentum breakout occurred

These engineered variables provide significantly richer information than the original OHLCV dataset and form the input feature set for machine learning models.

5.8 Final Dataset Verification

After preprocessing, several validation checks were performed to ensure dataset quality.

The processed dataset was verified for:

- Missing values.
- Duplicate observations.
- Invalid numerical values.
- Correct chronological ordering.
- Correct feature generation.
- Consistent company identifiers.
- Proper normalization of selected variables.

Only after successfully passing these validation checks was the dataset considered suitable for exploratory analysis and machine learning.

6. Exploratory Data Analysis ®

6.1 Environment Setup

Software Used: RStudio

Libraries Used:

- ggplot2
- Corrplot

```
install.packages("ggplot2")

install.packages("corrplot")


library(ggplot2)

library(corrplot)


stock_data$Date <- as.Date(stock_data$Date)


stock_data <- read.csv("processed_dataset.csv")

head(stock_data)

str(stock_data)
```

6.2 Visualization 1: Closing Price Over Time

R Code

```
ggplot(stock_data, aes(x = Date, y = Close)) +

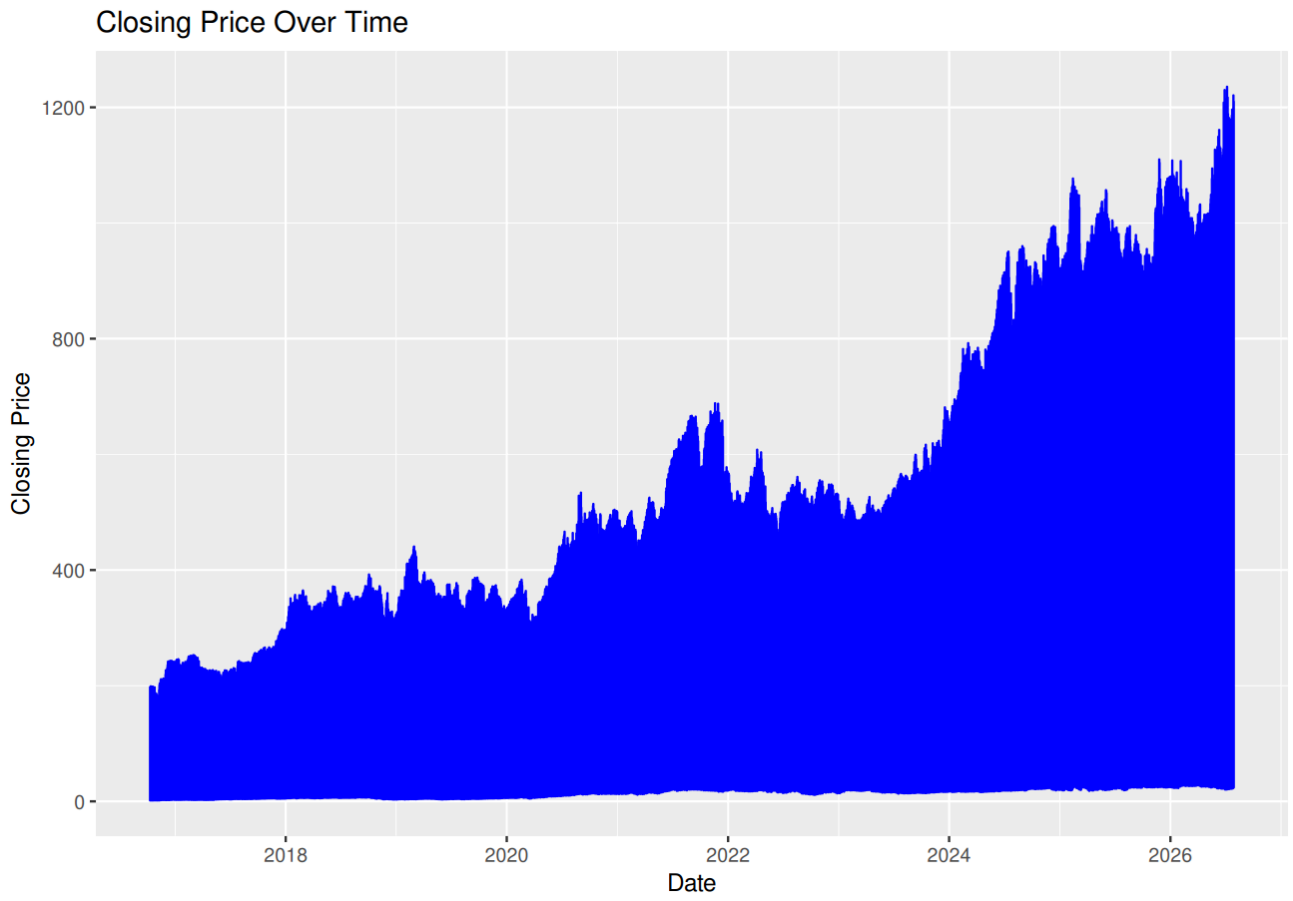
 geom_line(color = "blue") +

 labs(title = "Closing Price Over Time",

      x = "Date",

      y = "Closing Price")
```

Output



Observation

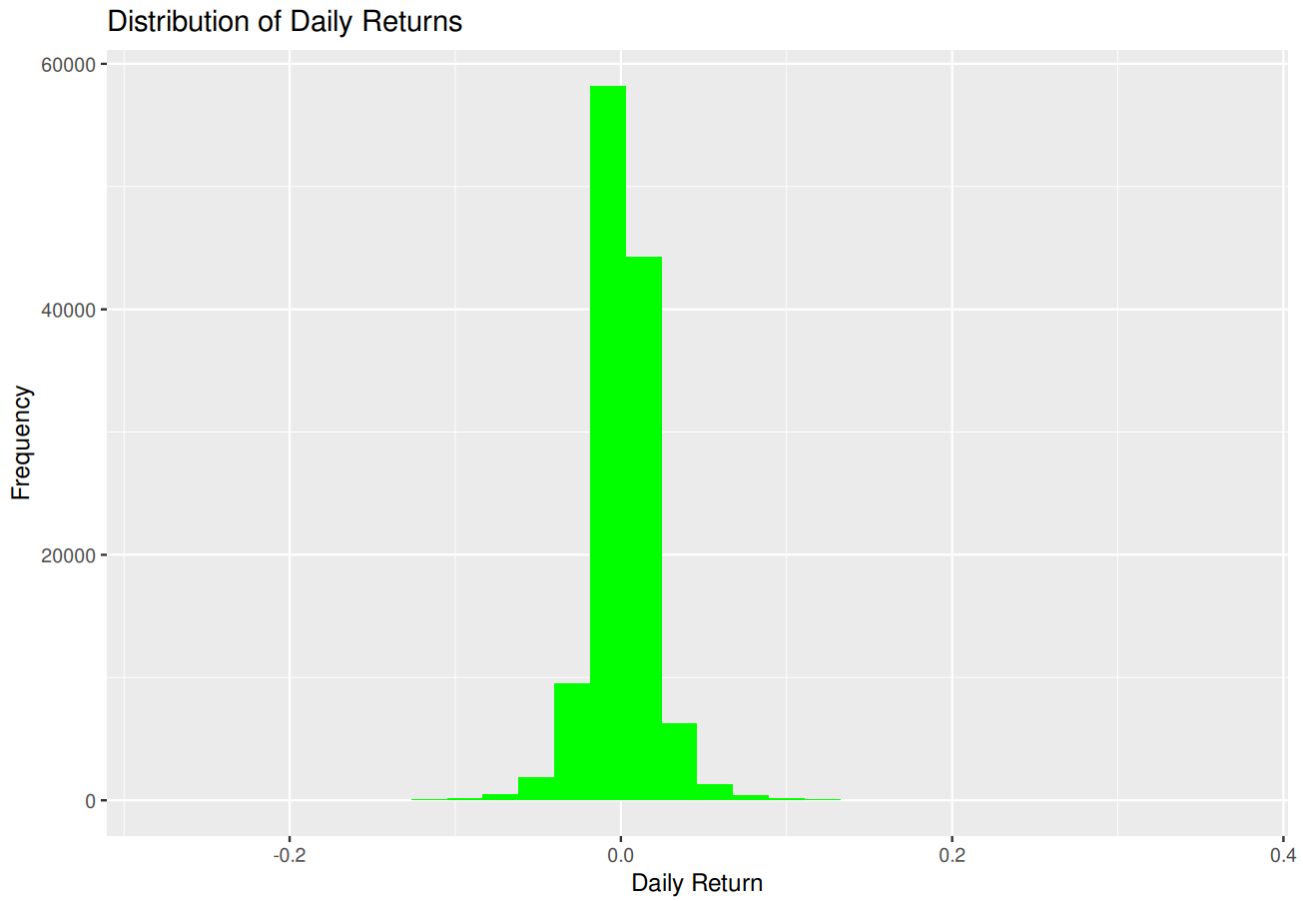
The line graph shows the historical trend of Apple's closing stock price over the selected period. The price exhibits long-term fluctuations with an overall upward trend, reflecting periods of market growth and temporary corrections.

6.3 Visualization 2: Distribution of Daily Returns

R Code

```
ggplot(stock_data, aes(x = Daily_Return)) +
  geom_histogram(fill = "green", bins = 30) +
  labs(title = "Distribution of Daily Returns",
       x = "Daily Return",
       y = "Frequency")
```

Output



Observation

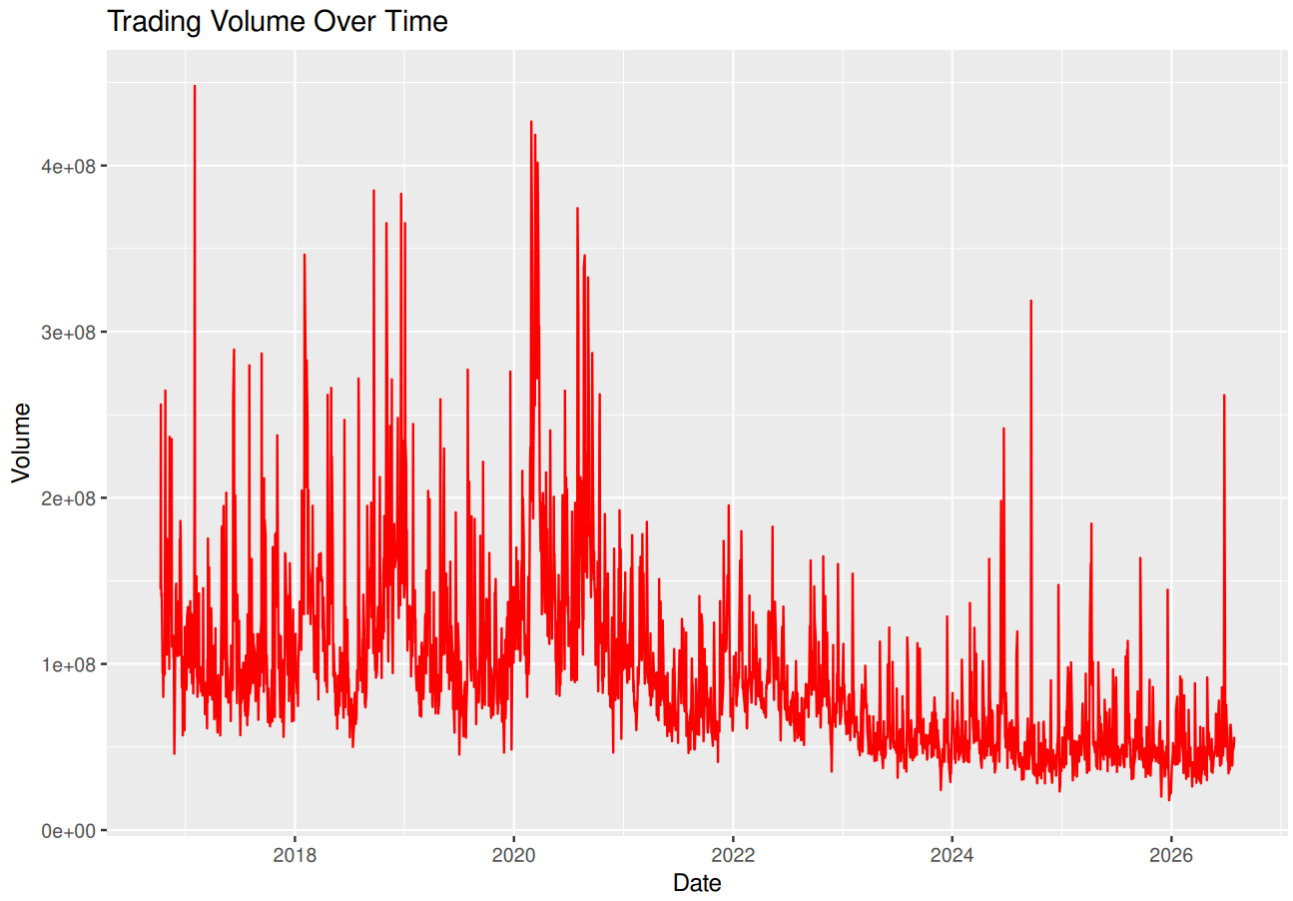
The histogram illustrates that most daily stock returns are concentrated around zero, indicating that small daily price changes occur more frequently than large fluctuations.

6.4 Visualization 3: Trading Volume Over Time

R Code

```
ggplot(apple, aes(x = Date, y = Volume)) +
  geom_line(color = "red") +
  labs(title = "Trading Volume Over Time",
       x = "Date",
       y = "Volume")
```

Output



Observation

The trading volume graph highlights periods of increased market participation. Peaks in trading volume generally correspond to significant market events or major price movements.

6.5 Visualization 4: Correlation Heatmap

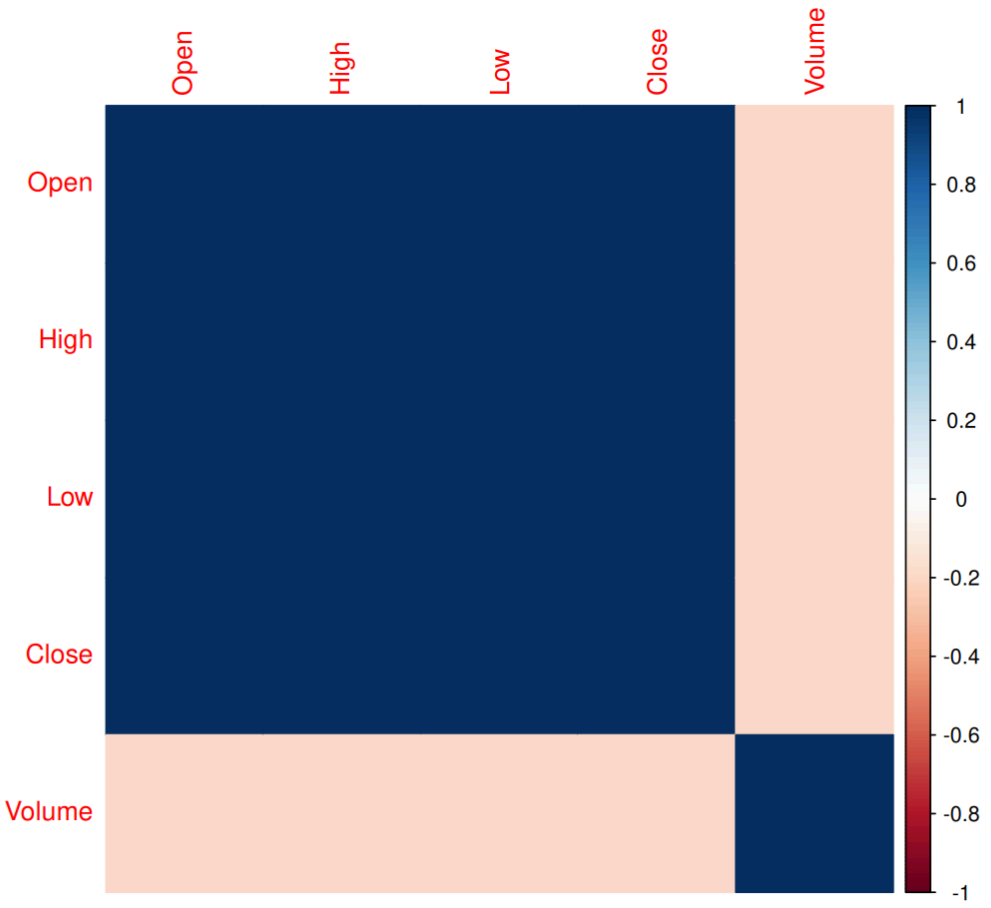
R Code

```
num_data <- stock_data[, c("Open", "High", "Low", "Close", "Volume")]

cor_matrix <- cor(num_data)

corrplot(cor_matrix, method = "color")
```

Output



Observation

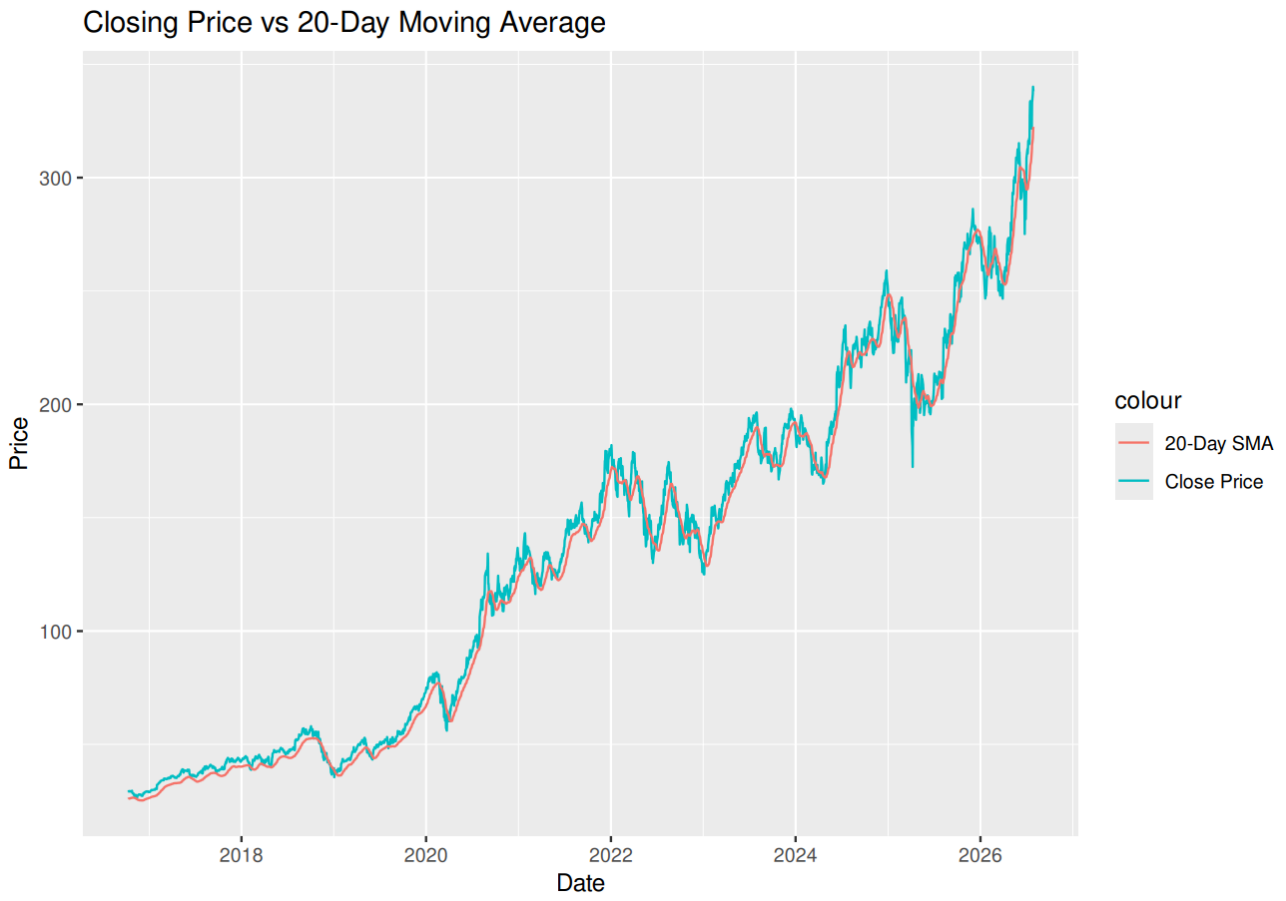
The heatmap indicates a strong positive correlation among the Open, High, Low, and Close price variables, while Volume shows a comparatively weaker relationship with price variables.

6.6 Visualization 5: Closing Price vs 20-Day Moving Average

R Code

```
ggplot(apple, aes(x = Date)) +
  geom_line(aes(y = Close, color = "Close Price")) +
  geom_line(aes(y = SMA_20, color = "20-Day SMA")) +
  labs(title = "Closing Price vs 20-Day Moving Average",
       x = "Date",
       y = "Price")
```

Output



7. Summary

This phase established a clean, well-justified, and diverse dataset of 50 S&P 500 stocks across 10 sectors spanning 10 years of daily trading history, engineered a set of literature-grounded technical indicators, defined a quantitative momentum-breakout target label, and validated initial patterns through exploratory visualization. The processed dataset (123,150 rows × 45 columns) is now normalized, encoded, and structured for the model-building phase, where classification algorithms (e.g. Random Forest, XGBoost, LSTM) will be trained to predict the Breakout and Trend_Binary target labels defined here.

References

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